

BADHRINARAYAN MALOLAN

Email: badhrinarayan@vt.edu | LinkedIn: linkedin.com/in/badhri-narayan | GitHub: github.com/badhri-n | Phone: (540) 558-5924

TECHNICAL SKILLS

Languages: Python, Java, MATLAB, SQL, Shell Scripting, JavaScript, Tableau, R
ML/AI Frameworks: PyTorch, TensorFlow, Keras, PyTorch Lightning, OpenCV, Scikit-learn, XGBoost, MLFlow
MLOps & DevOps: Git, Docker, Kubernetes, CI/CD, Azure Cloud, Maven, Jira, Apache Kafka, Slurm

EXPERIENCE

Wireless @ Virginia Tech

Data Scientist - Researcher

Aug 2023 – present

Blacksburg, VA

- Built CNN Autoencoders in Tensorflow for QPSK signal interference suppression, achieving 30dB improvement over traditional methods & reducing MSE by >200%. Used model training pipelines with automated hyperparameter tuning.
- Developed Cyclostationary phase-aligned models combining signal processing & Deep Learning (DL), achieving 95% parameter reduction with similar performance. Led model compression & parallelization initiatives.
- Implemented data preprocessing pipelines & adversarial training (FGSM, PGD), reducing MSE by 18% & modulation misclassification by 16%. Enhanced reconstruction accuracy through data augmentation workflows.
- Used t-SNE, UMAP for analysis of clustered signal data & XGBoost for feature importance analysis. Conducted cross-SNR A/B testing and model generalizability analysis to audit performance in various scenarios.
- Exploring multimodal Attention based architectures & Transformers for signal representation, achieving 12% improvement. Investigating fine-tuning pre-trained LLMs for next generation open radio units (oRU).

Oracle Financial Services Software (OFSS) Ltd.

Software Developer - ML

Oct 2020 – Jun 2022

Bengaluru, India

- Built real-time streaming infrastructure with Apache Kafka & Java, replacing RabbitMQ. Implemented CI/CD pipeline on Oracle Cloud, reducing data transfer latency by 30% for ML training & API optimization.
- Led ML experiments with Pandas EDA & deployed K-means, Random Forest, Isolation Forest models for AML anomaly detection, increasing fraud detection rate by 22% over baseline methods in production pipelines.
- Built automated data mapping system using NLP text similarity algorithms (NLTK, SpaCy) to transform customer audit data into internal formats, improving data pipeline efficiency by 65% for reporting workflows.
- Designed data processing pipelines with SpaCy for XML & JSON parsing. Optimized field extraction using rule-based & ML approaches, reducing processing time by 30% over manual processing.

Centre of Excellence in Complex and Nonlinear Dynamical Systems (CoE-CNDS), VJTI

Research Assistant

Aug 2019 – Aug 2020

Mumbai, India

- Built multimodal DL pipelines for synthetic Deep-Fake detection across visual & audio domains using HPC (NVIDIA-DGX1, IBM AC922), achieving 93.4% accuracy on diverse datasets with optimized training workflows.
- Applied Transfer Learning in XceptionNet with Explainable AI (XAI) for model interpretability. Published at ICICT 2020 & achieved Bronze medal in DFDC Kaggle challenge through large-scale experimentation methodologies.

Tata Consultancy Services (TCS) Ltd.

Project Trainee- Computer Vision Engineer

Jun 2019 – Jul 2020

Mumbai, India

- Built real-time hand detection system by fine-tuning pre-trained SSD models, achieving 0.45 IoU. Implemented optimized inference pipeline with 92% speed improvement via multi-threading for industrial safety monitoring.

PUBLICATIONS

“An End-to-End CNN-based Interference Suppression Autoencoder for Wireless Communications”, Under Review

“Deep Learning Interference Suppression for Cyclostationary Signals”, IEEE MILCOM 2025

“Explainable Deep-Fake Detection Using Visual Interpretability Methods”, IEEE ICICT 2020

KEY PROJECTS

PhotoBoost: Image Transformation | TensorFlow, PyTorch, JavaScript Next.js, HTML, Django

Created an AI-driven Image Processing (IP) application using Agile methodology. Developed a Next.js web app & fine-tuned DL models for advanced editing features- image stitching, colorization, super-resolution, denoising, style transfer

PolicyAdvisor: Government Policy Research RAG | Python, PostgreSQL, FAISS, Transformers

Built a RAG system for government policy research using sentence transformers and FAISS for semantic search across federal documents with PostgreSQL for metadata management.

Augmented CNNs for Remote Sensing Change Detection | TensorFlow, PyTorch, NumPy, SciPy

Enhanced Remote Sensing Change Detection algorithms via Siamese CNNs, implementing Edge features, Multiresolution & Feature augmentations. Achieved 4% IoU improvement and robustness to Dataset Shift, Noise & few-shot learning scenarios.

EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech)

Master of Science (Thesis), Computer Engineering (GPA: 3.96/4)

Blacksburg, VA

Aug 2022 – Feb 2025

Veermata Jijabai Technological Institute (VJTI)

Bachelor of Technology (B.Tech), Electronics & Telecommunication Engineering (CPI: 8.24/10)

Mumbai, India

Jul 2016 – Sept 2020